SOC PRACTICAL APPLICATION REPORT

Document Version: 1.0

Document Information

Item	Details
Assessment Type	SOC Monitoring and Brute-Force Detection using SIEM
Tools Used	Windows Event Viewer, Splunk, Wazuh SIEM, Elastic SIEM, LogParser Lizard
Target Scope	Authorized Virtual Machines (Windows VM and Ubuntu Endpoint)
Report Prepared By	Pal Piyushkumar Patel
Date	26-02-2026

Introduction

This document presents a practical implementation of Security Operations Center (SOC) monitoring activities conducted in a simulated lab environment. The purpose of this documentation is to demonstrate how security logs, dashboards, and SIEM tools can be used to detect and analyze brute-force login attempts and unauthorized access activities.

This report is intended for:

- Internal reference for SOC training
- Academic evaluation
- Cybersecurity lab documentation

Visuals such as dashboards, event logs, and SIEM alerts are included to enhance understanding of complex detection workflows and monitoring configurations.

1. Log Analysis

Windows Event Viewer was used to analyze system security logs for detecting suspicious login activities.

The following Event IDs were filtered:

- **Event ID 4625** – Failed Login Attempt
- **Event ID 7045** – New Service Creation

These logs were examined to identify patterns associated with brute-force login attempts from Windows Security Logs.

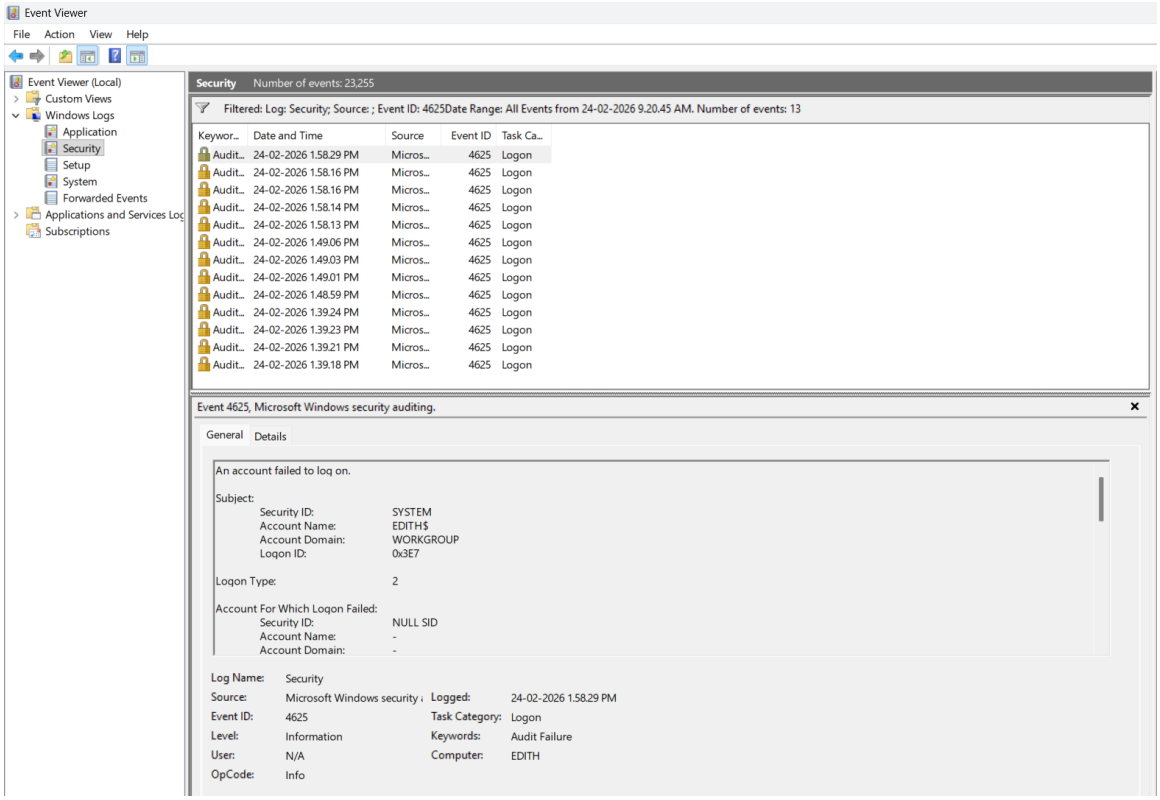
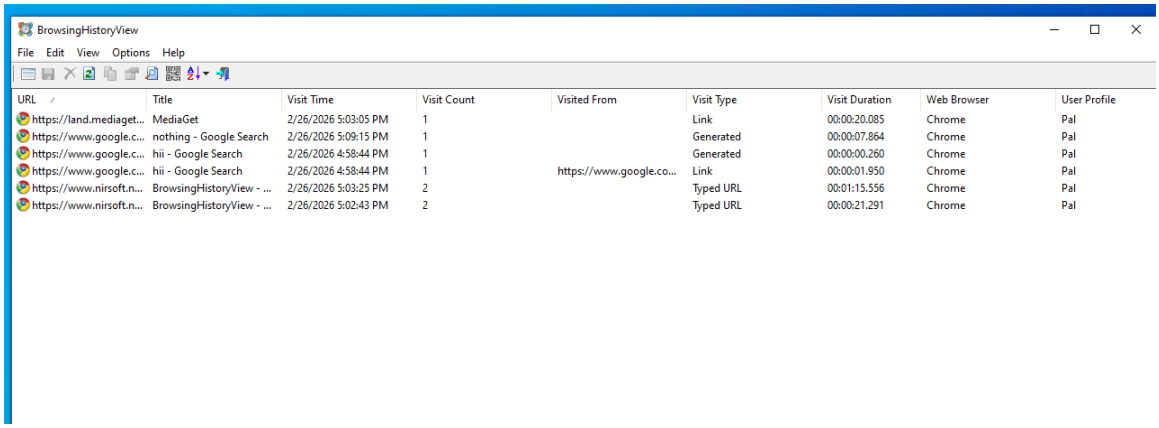


Figure 1: Failed Login Detection using Windows Event Viewer (Event ID 4625)
Source: Self-generated using Windows Event Viewer Lab

1.1 Browser History Analysis

Browser history artifacts were analyzed using *Eric Zimmerman's LECmd Tool* to detect visits to a test URL.



URL	Title	Visit Time	Visit Count	Visited From	Visit Type	Visit Duration	Web Browser	User Profile
https://land.mediaget...	MediaGet	2/26/2026 5:03:05 PM	1		Link	00:00:20.085	Chrome	Pal
https://www.google.c...	nothing - Google Search	2/26/2026 5:09:15 PM	1		Generated	00:00:07.864	Chrome	Pal
https://www.google.c...	hii - Google Search	2/26/2026 4:58:44 PM	1		Generated	00:00:00.260	Chrome	Pal
https://www.google.c...	hii - Google Search	2/26/2026 4:58:44 PM	1	https://www.google.co...	Link	00:00:01.950	Chrome	Pal
https://www.nirsoft.n...	BrowsingHistoryView - ...	2/26/2026 5:03:25 PM	2		Typed URL	00:01:15.556	Chrome	Pal
https://www.nirsoft.n...	BrowsingHistoryView - ...	2/26/2026 5:02:43 PM	2		Typed URL	00:00:21.291	Chrome	Pal

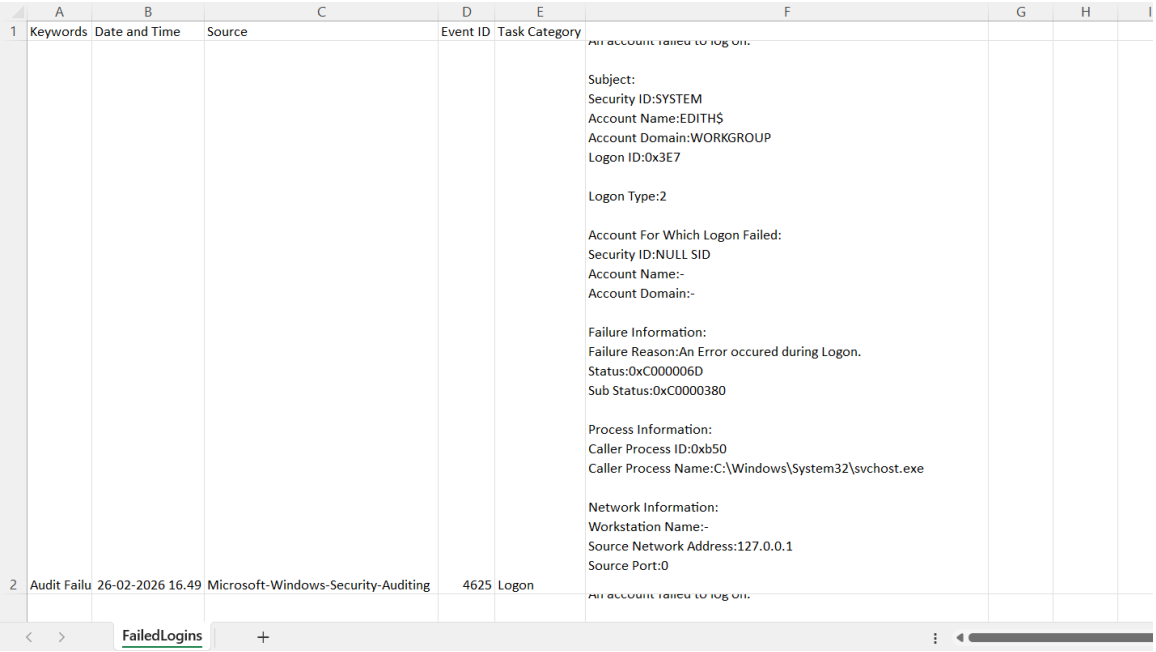
Figure 2: Browser History Analysis using LECmd Tool
Source: Self-generated using LECmd Tool

1.2 Advanced Task: Brute Force Detection

Failed login attempts were generated in a Windows Virtual Machine using repeated incorrect password entries.

The security logs were:

1. Filtered using Event Viewer for **Event ID 4625**
2. Exported into CSV format
3. Analyzed for brute-force attack indicators



	A	B	C	D	E	F
1	Keywords	Date and Time	Source	Event ID	Task Category	
2	Audit Failu	26-02-2026 16.49	Microsoft-Windows-Security-Auditing	4625	Logon	An account failed to log on. Subject: Security ID:SYSTEM Account Name:EDITH$ Account Domain:WORKGROUP Logon ID:0x3E7 Logon Type:2 Account For Which Logon Failed: Security ID:NULL SID Account Name:- Account Domain:- Failure Information: Failure Reason:An Error occured during Logon. Status:0xC000006D Sub Status:0xC0000380 Process Information: Caller Process ID:0xb50 Caller Process Name:C:\Windows\System32\svchost.exe Network Information: Workstation Name:- Source Network Address:127.0.0.1 Source Port:0

FailedLogins

Figure 3: Exported Failed Login Events (CSV Format)
Source: Windows Security Logs

Tools Used

Built-in Tools:

- Windows Event Viewer
- wevtutil CLI

Third-party Tools:

- LogParser Lizard

2. Security Event Documentation

Security events were recorded using a structured SOC documentation template to maintain consistency in log tracking and incident analysis.
Security events were documented using the following template:

Date/Time	Source IP	Event ID	Description	Action Taken
25-02-2026	192.168.1.43	4625	Multiple Failed Logins	Alert Generated

3. Monitoring Dashboard Setup

Monitoring dashboards were created in Splunk to visualize critical security data such as:

- Top 10 Source IPs generating alerts
- Frequency of critical Event IDs

Custom Splunk rules were configured for SOC monitoring to provide real-time alerting.

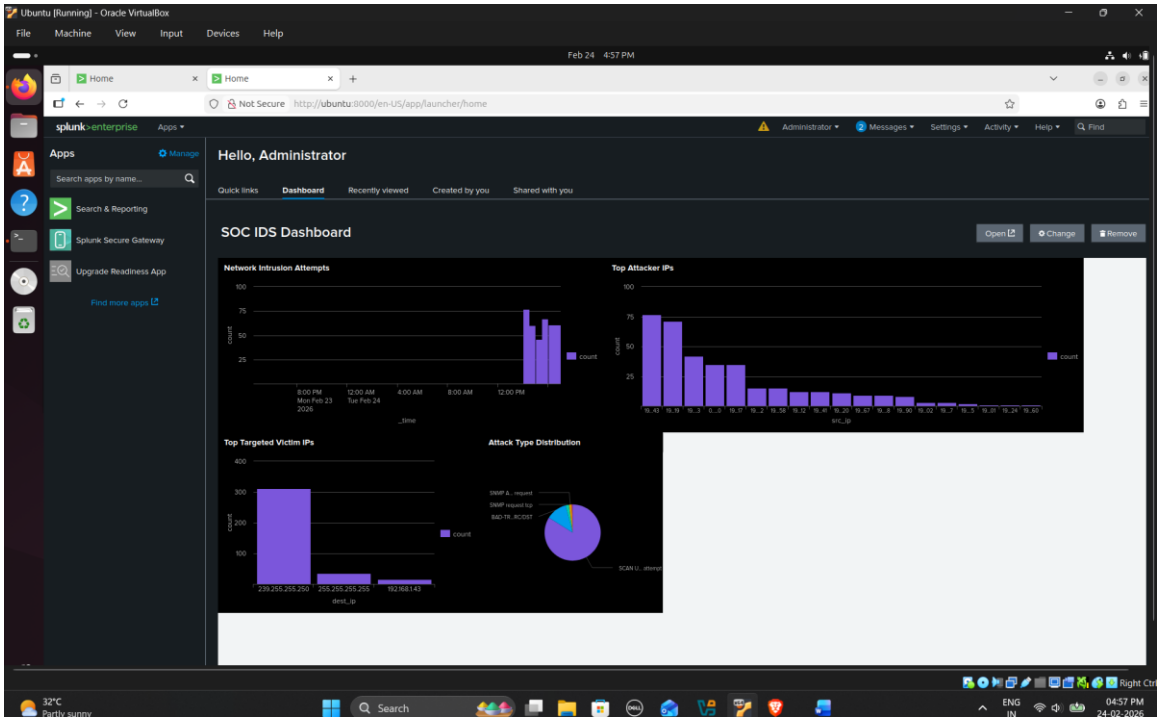


Figure 4: Monitoring Dashboard Visualization
Source: Splunk Dashboard

4. Configure Alert Rules

4.1 Splunk Alert Rules Configuration

Snort Rule Created: Detect 5+ failed logins in 5 minutes

Index: security-login-*

Condition: count > 5

Figure 5: Splunk Alert Rule Configuration

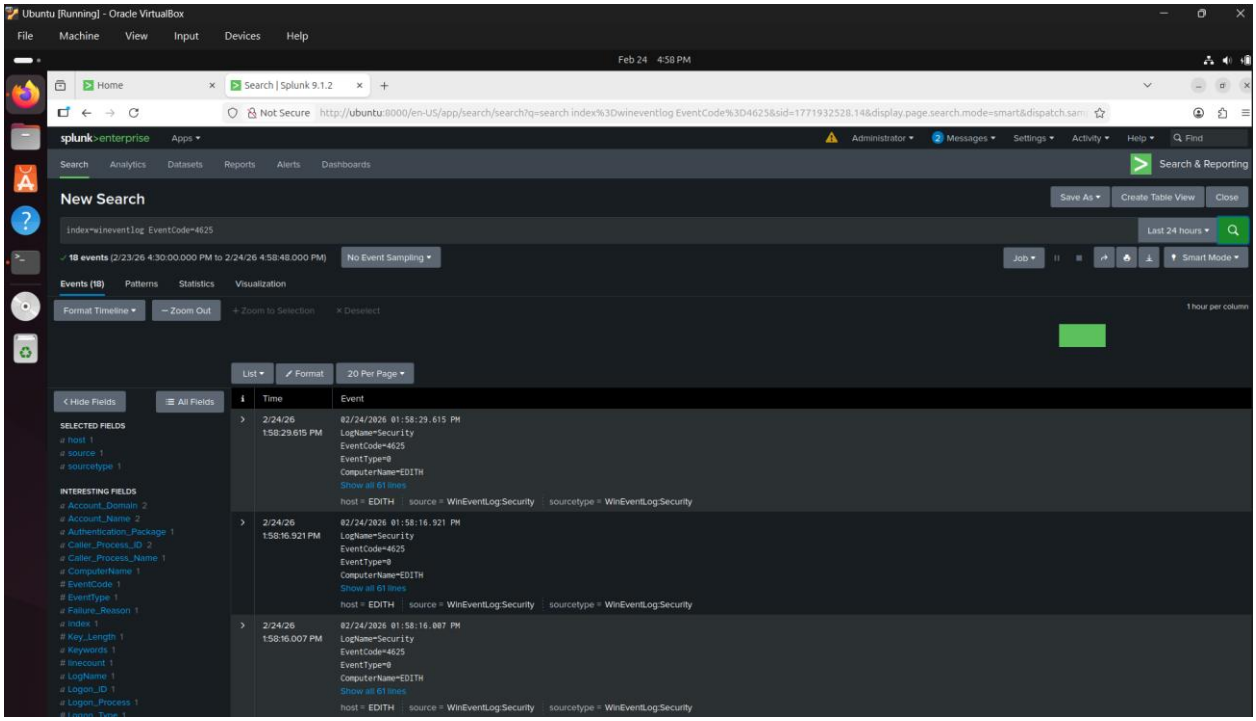


Image Source: Splunk Query Interface

4.2 Advanced Task: SSH Brute Force Detection using Wazuh SIEM

A simulated SSH brute-force attack was performed from a Windows host to an Ubuntu server using invalid login credentials.

The Wazuh SIEM platform was configured to monitor authentication logs from:

- /var/log/auth.log (Ubuntu Endpoint)

Repeated unauthorized login attempts generated security events which were successfully detected by the built-in Wazuh rule.

The generated alert details are as follows:

- **Rule ID:** 5710
- **Description:** sshd: Attempt to login using a non-existent user
- **Agent Name:** ubuntu
- **Severity Level:** 5
- **Source IP:** 192.168.147.129

This indicates multiple unauthorized SSH login attempts which may represent brute-force credential attacks.

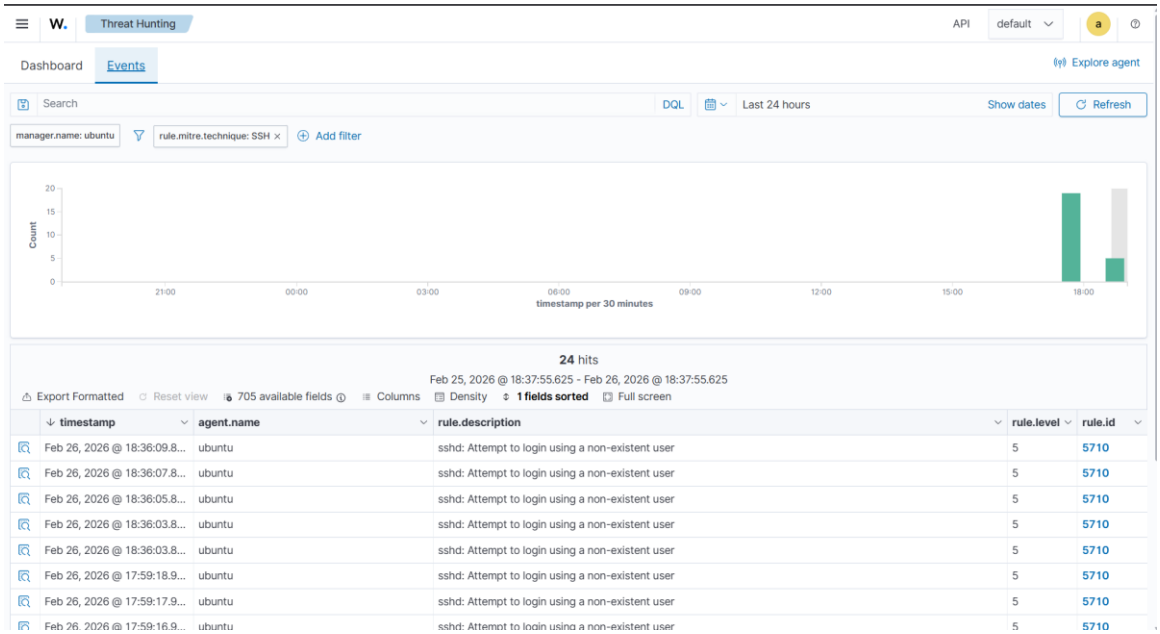


Figure 5: SSH Failed Login Attempts Detected in Wazuh Dashboard
Source: Wazuh SIEM Dashboard

5. MITRE ATT&CK Mapping

The generated alert was analyzed using the integrated MITRE ATT&CK framework within the Wazuh Dashboard.

The detected activity was mapped to the following adversarial technique:

- **Tactic:** Credential Access
- **Technique:** T1110 – Brute Force

This mapping indicates that repeated login failures may be associated with brute-force attacks attempting to gain unauthorized system access.

213 hits

Feb 25, 2026 @ 18:41:31.145 - Feb 26, 2026 @ 18:41:31.145

Export Formatted | Reset view | 705 available fields | Columns | Density | **1 fields sorted** | Full screen

timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Feb 26, 2026 @ 18:36:32.5...	Windows-vm	T1484	Defense Evasion, Privilege Escalation	Special privileges assigned to new logon.	3	67028
Feb 26, 2026 @ 18:36:32.5...	Windows-vm	T1078	Defense Evasion, Persistence, Privileg...	Windows Workstation Logon Success	3	60118
Feb 26, 2026 @ 18:36:32.4...	Windows-vm	T1078	Defense Evasion, Persistence, Privileg...	Windows Workstation Logon Success	3	60118
Feb 26, 2026 @ 18:36:09.8...	ubuntu	T1110.001 T1	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:07.8...	ubuntu	T1110.001 T1	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:05.8...	ubuntu	T1110.001	Credential Access	PAM: User login failed.	5	5503
Feb 26, 2026 @ 18:36:05.8...	ubuntu	T1110.001 T1	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:03.8...	ubuntu	T1110.001 T1	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:36:03.8...	ubuntu	T1110.001 T1	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
Feb 26, 2026 @ 18:35:07.8...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:33:13.8...	ubuntu	T1548.003	Privilege Escalation, Defense Evasion	First time user executed sudo.	4	5403
Feb 26, 2026 @ 18:33:13.8...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:29:03.4...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:29:03.2...	ubuntu	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login session opened.	3	5501
Feb 26, 2026 @ 18:28:29.4...	ubuntu	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402

Figure 6: MITRE ATT&CK Mapping for SSH Brute Force Attempt
Source: Wazuh Dashboard

Conclusion

The SOC lab environment successfully simulated real-time monitoring, alert generation, brute-force detection, and endpoint monitoring using SIEM tools. The practical implementation demonstrated how log analysis and alert configuration can be leveraged to detect credential-based attacks in enterprise environments.

References

- Splunk Documentation
- Wazuh Documentation
- MITRE ATT&CK Framework